

a) On what intervals is the graph of f increasing? (2)

$(1, 2)$ where $f' > 0$

1.5

b) On what intervals is the graph of f concave down? (2)

~~$(-1, 0.1)$~~ $(-1, 0.1)$ why?
 ~~$(2.5, 4)$~~ $(2.5, 4]$ why?

c) Determine all points where the graph of f has local extrema. $1.5/(2)$

$x=1$ and $x=3$ why?
 Local min Local max

d) Determine the x-coordinate of the inflection point(s) of the graph of f $1.5/(2)$

$x=-1$ why
 $x=0.1$
 $x=2.5$

just dots as interval is closed

e) Sketch a possible graph of f on $[-3, 4]$. You can draw it on the same graph or on the separate graph.

1.5
 (2)

$$y' = 3x^2 \frac{d}{dx} \sqrt{5-x} + \sqrt{5-x} \cdot 6x$$

$$y' = 3x^2 \left(\frac{-1}{2\sqrt{5-x}} \right) + \sqrt{5-x} \cdot 6x$$

$$0 = 3x^2 \left(\frac{-1}{2} \right) + (5-x)(6x)$$

$$0 = -7.5x^2 + 30x$$

$$= -7.5(x-4) \cdot x$$

2. Determine absolute extreme values for $f(x) = -\frac{3}{4}x^4 - 2x^3 + 12x^2 - 6$ in $[-5, 3]$

$$f'(x) = -3x^3 - 6x^2 + 24x$$

$$= -3x(x^2 + 2x - 8)$$

$$f'(x) = -3x(x+4)(x-2)$$

$$f'(x) = 0 \text{ at } 0, -4, 2$$

(x)

$$f(0) = -6$$

$$f(-4) = 122$$

$$f(2) = 14$$

$$f(-5) = -424.75$$

$$f(3) = -12.75$$

Abs min $f(-5) = -424.75$

Abs max

$$(-4, 122)$$

Abs min

$$(-5, -424.75)$$

3. State whether or not the function satisfies the hypothesis of the Mean Value Theorem on the given interval, and if it does, find the value of c in the interval (a, b) guaranteed by the theorem:

a) $f(x) = x^3 - x$ on $[0, 2]$

Differentiable as

$f(x)$ cont over $[0, 2]$ and

diff over $(0, 2)$

$$f'(x) = 3x^2 - 1$$

$$= \frac{f(2) - f(0)}{2 - 0}$$

$$= \frac{6 - 0}{2 - 0} = 3$$

$$3 = 3x^2 - 1 \Rightarrow x^2 = \frac{4}{3} \Rightarrow x = \pm \sqrt{\frac{4}{3}}$$

$$x = \pm \sqrt{\frac{4}{3}} \Rightarrow x = \pm \frac{2}{\sqrt{3}}$$

4. Given $f(x) = \frac{(\ln x)^2}{2}$, determine intervals of increase and decrease. Study concavity and determine any point of inflection.

$$f(x) = \frac{(\ln(x))^2}{2} \quad x > 0$$

$$f'(x) = \frac{1}{2} (2 \ln(x)) \cdot \frac{1}{x}$$

$$= \frac{1}{x} \cdot (\ln(x)) \cdot \frac{1}{x}$$

$$f'(x) = \frac{\ln(x)}{x}$$

$$\ln(x) = 0 \Rightarrow x = 1$$

$$x < 1$$

$$x > 1$$

b) $y = (x-3)^{-2}$ in $[1, 4]$

3.5/4

Not continuous over $[1, 4]$ so

mean value theorem does not

$$f(3) = \frac{1}{(3-3)^2} = \frac{1}{0}$$

Guaranteed is $c = \frac{2}{\sqrt{3}}$

$c = \frac{2}{\sqrt{3}}$ is when slope is 3

$$c = \pm \sqrt{\frac{4}{3}}$$

determine any point of inflection.

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